

5 Long Hole Raise

10/10/2019

Applicability I

- *Both level, upper level and lower are to be available for raising.
- *It is possible to excavate raise whose angle varies from 90 degree to 50 degree.
- *Length of raise is a limited to 40m.
- *Accurate drilling and proper blasting is required for success of method.

Applicability I

- *Disturbed ground with joints etc are not suitable as frequent jamming of drill rods and bit takes place.
- *Long hole raising technique is applied for excavating slot raise for Sublevel Stoping & Sublevel Caving method.
- *It is not possible to support raise wall at time of excavation.

Drilling Preparation I

- *Long holes are to be drilled from upper level to lower level in cylindrical or burn cut pattern.
- *Hole length & inclination is kept same as that of intendent raise to be driven.
- *For drilling long holes, pneumatic or hydraulic drifter is used.
- **Drilling long holes are done by Down The Hole (DTH) or In The Hole (ITH) pneumatic drill machines.
- *Pneumatic drill machines are normally mounted on vertical columns and horizontal bar structure.

Drilling Preparation II

- *Pneumatic drifter mounted on bar & column arrangement requires additional 0.5m space all around to install & operate machines.
- *For pneumatic machines, set up has to be dismantled, shifted & assembled manually every time.
- *This process is labour intensive & time consuming.
- *Hydraulic drifter comes mounted on drill rig.
- *Drill rig can be self propelled or shifted with help of service vehicle.
- *Drill rig mounted hydraulic drifter requires minimum 1m additional space all around for operation.

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Drilling Preparation III

- *Site to be checked by driller & supervisor to decide up on exact location of drilling set up, considering different parameter like height, clearance, drive condition or any other parameter effecting installation & operation.
- *Floor to be checked up by blaster to ensure there is no misfire.
- *All loose rock is to be removed from floor & a horizontal layer of PCC is layed for better machine ibstallation.
- *All service line like power, compressed air, water, ventilation etc are to be brought to site.

Drilling Preparation IV

- *In case of column & bar mounting arrangement is used, jacks are to be tightened properly before drilling starts.
- *Drill pattern followed is either burn cut or cylindrical cut.
- *In cylindrical cut, few center holes of larger diameter, up to 100mm are required to be drilled.
- *Larger diameter holes are achieved by rearming normal size blast hole.

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Drilling Preparation V

- *Hole drilling sequence should be such that, minimum time should be required to shift drifter from one hole to another hole.
- *At time of drilling, drill machine is set to drill at an angle at which raise has to be driven.
- *Accessories required at drilling site are Drill bit, collaring bit, coupling, adopter, extension drill steel, drill rod, bit, spanners, hole collaring pipe, tapered wooden wedges, crow bars, spades etc.

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Drilling Preparation VI

*For measuring angle, instruments “Bruton” is used.

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Survey I

- *Place of starting point for raise (lower level) has to be decided after through survey of starting point of raise and punching point of raise (upper level).
- *Surveyor has to mark center point of raise, raise boundary, intended direction at drill site.
- *Drill hole pattern is to be marked at back of raise face & PCC floor.

Survey I

- *In case of incline raise, inclination of each hole has to be calculated.
- *Before drilling holes, some amount of corrective angle has to be provided to reduce total deviation.
- *Drill holes are to be surveyed for their accuracy.
- *Deviation of each hole should not be more than 2 %.

Survey I

- *Immediately after drilling holes, holes are to be surveyed.
- *In case, deviation exceeds tolerance limit (not exceeding 1-2%), additional holes are to be drilled.

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Drilling 1

- *Central hole is drilled first.
- *Central hole is of larger diameter and serves as relieve hole.
- *Each hole is collared for initial 0.5mn drilled by larger dia bit.
- *PVC pipe is used for protecting drill hole collar.
- *Drill hole diameter varies from 57mm to 115mm.
- *Each hole is collared for initial 0.5m using a little larger diameter bit, so that PVC stand pipe can be fixed.

Drilling II

- *For 57mm diameter hole, 65mm diameter bit can be used for PVCF stand pipe as casing.
- *Drill machine should run at low RPM to reduce deviation of hole.
- *Drill holes should be flushed at every 0.5m progress.
- *This shall help to avoid bit & rod jamming in hole & extend bit life by avoiding regrinding of cuttings.
- *Once holes are through to lower level, holes are flushed out by water.

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Drilling III

- *It is very important to drill hole with accuracy, without deviation.
- *To achieve drilling accuracy, it is important to maintain drill machine, its components, jack, clamp in good order & tighten before starting if drilling, in between drilling.
- *List of accessories to be maintained at drill site.
- *Bit for normal drilling, 57mm dia.
- *Bit for collaring. 64mm diameter.
- *Reamer bit 104mm diameter.
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Drilling IV

- *Coupling of different sizes.
- *Clamps of different sizes.
- *Adapters.
- *Extension drill steel.
- *Rod & bit spanners.
- *Lubricants.
- *PVC stand pipe.
- *Tapper wooden plug.
- *Crow bars.
- *Spades.
- *Picks etc.

DTH Machine I

- *In down the hole (DTH) drilling. Drill machine is in drill hole it self.
- *Drill machine is placed immediately after the bit.
- *In this case no energy is dissipating through drill steel or coupling.
- *Drill hole penetration rate is almost constant regardless depth of hole.
- *Drill machine is pneumatically operated & flushing is done by compressed air with water mist injection.

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DTH Machine II

- *In some cases, dry drilling with dust collector becomes essential to avoid use of water.
- *DTH machine is not designed to stand large quantity of water for flushing.
- *Most DTH drill machines are operated at 1.7mPa (18kg/cm², 250psi).

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DTH Machine III

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Drill Hole Plugging I

- *All blast holes are through to level below.
- *Holes are to be plugged before charging with explosive.
- *This process is done with help of wooden plug, polyethene bag, hessian cloth, old jute bag.
- *First, plugging material is tied by wire or rope & lowered down hole, till it gets through to lower level.

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Drill Hole Plugging II

- *Plugging material so suspended, is pulled up slowly, while rock cutting is poured from top of hole simultaneously.
- *This action plugs bottom of hole properly.
- *Rope / wire should be pulled tight & tied to anchor bolt fixed at raise top to ensure proper plugging of holes.
- *This is skilled job & should be tried out by trained persons only.
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Blasting I

- *After plugging, holes are charged with explosive, equal to length of round that usually 2.5 to 3m, while keeping rest of hole empty.
- *Hole is stemmed with rock cutting / chips or some stemming material poured in hole.
- *In non watery condition, use of explosive ANFO with suitable primer is used.
- *ANFO is cheapest explosive.
- *ANFO is poured in to hole with gravity.
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Blasting II

- *In Indian condition, Emulsion explosive is widely used for blasting.
- *Use of delay detonator is made to achieve sequential blasting.
- *Before every charge, all holes should be cleared & made empty.
- *If some hole is found to be jammed due to previous blasting & can not be cleared through by compressed air, then hole can be cleared by mild blasting of neighboring hole.

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Blasting III

- *Before blasting, workers at both levels should be warned & approaches to raise side (at both levels) should be well guarded.
- *Special care has to be taken, when parting to get through upper level is 5m or less.
- *This portion has to be taken in such a way, so that excessive loose or crater formation does not take place.

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Blasting IV

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- *Special care has to be taken when parting to get through into upper level is 5 m or less.
- *At end of raising, explosive load is reduced to prevent crater formation at raise mouth at upper level.
- *Blasting is based on Vertical Crater Retreat (VCR) concept.
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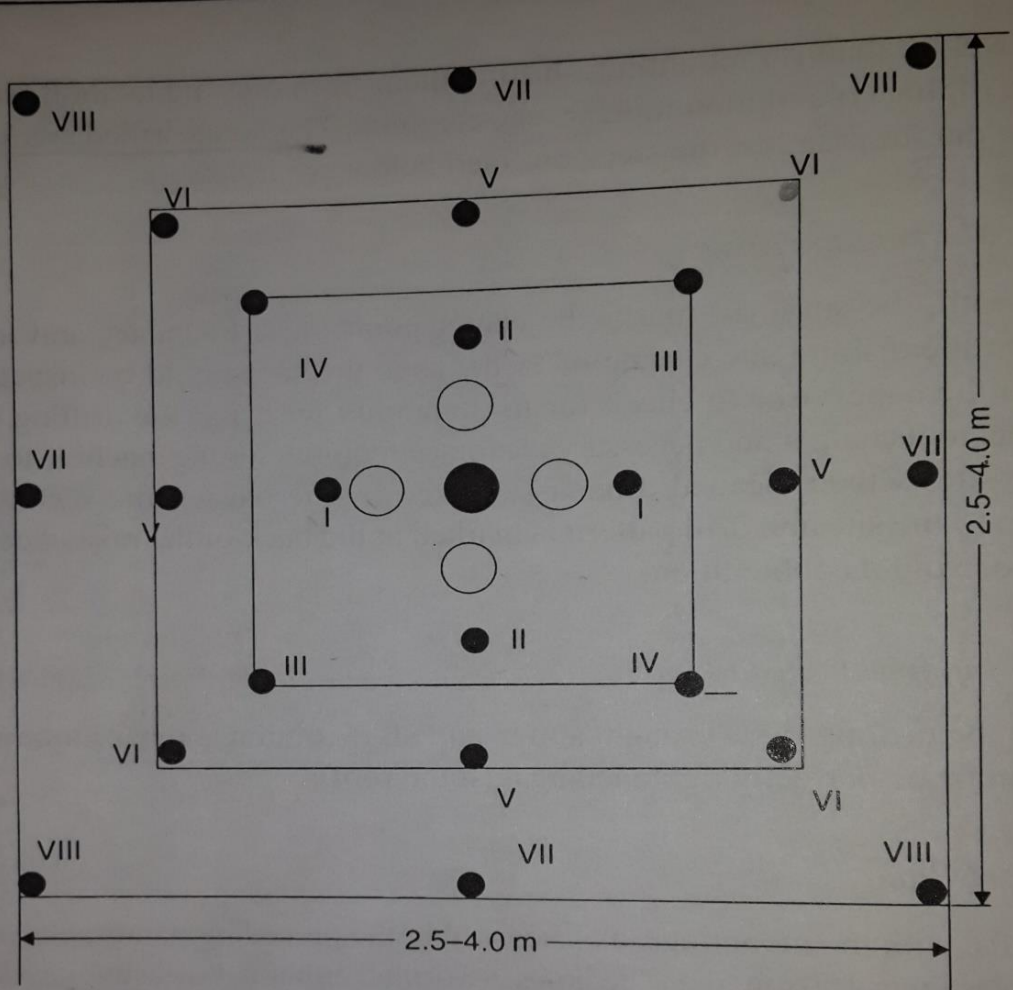
Blasting V

- *Crater in blasting technology is applied for creation of surface cavity in rock mass (VCR).
- *Explosive charges in crater theory are spherical or its geometric equivalent (L/D ratio 1 : 4 min, 6 max)

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Raise Long Hole

*Drill hole pattern

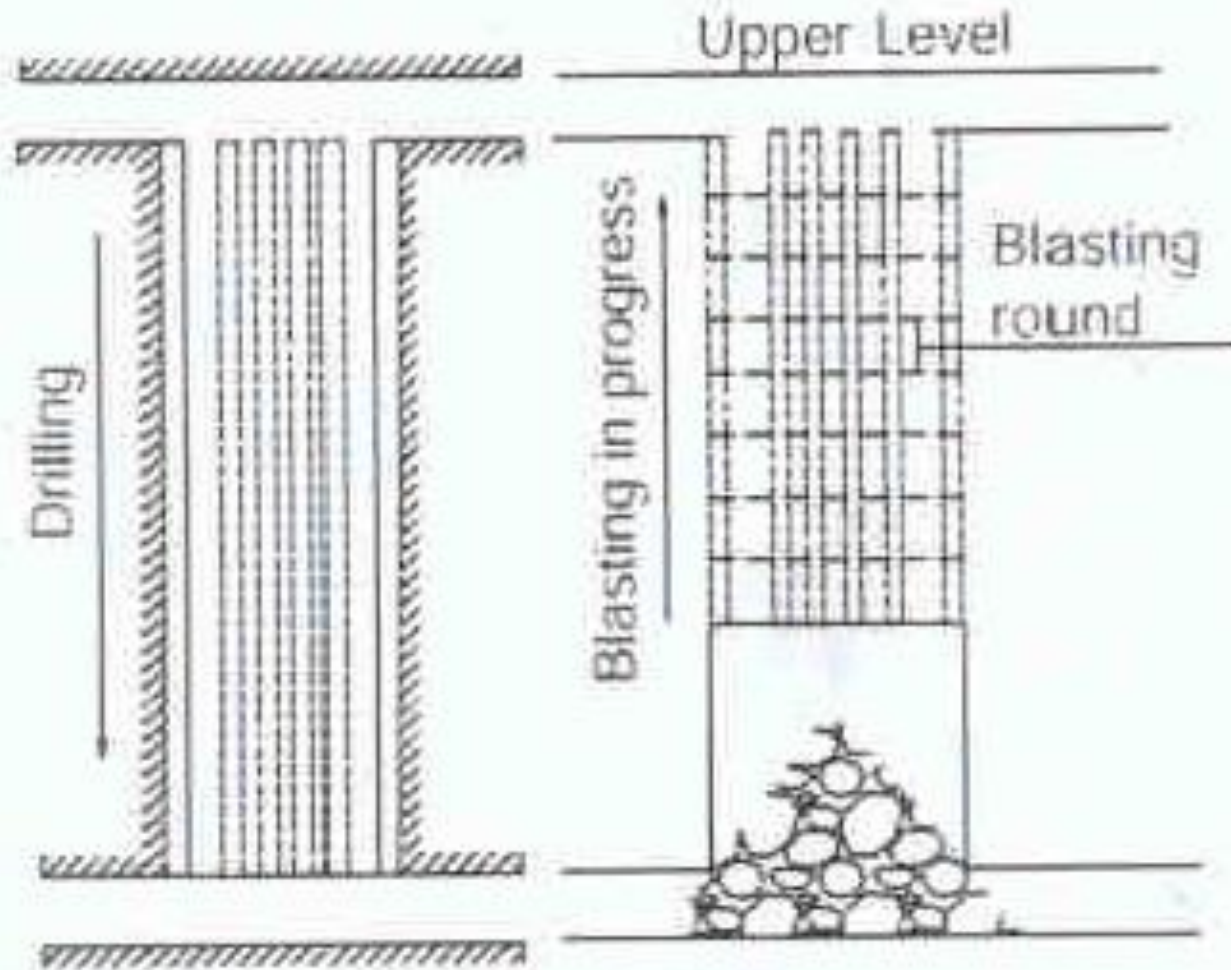


Legend

- - Charged holes 50-55 mm dia.
- - Uncharged holes 75-100 mm dia.
- - Charged holes 75-100 mm dia.
- Roman letters indicate delay numbers

Long Hole Raise I

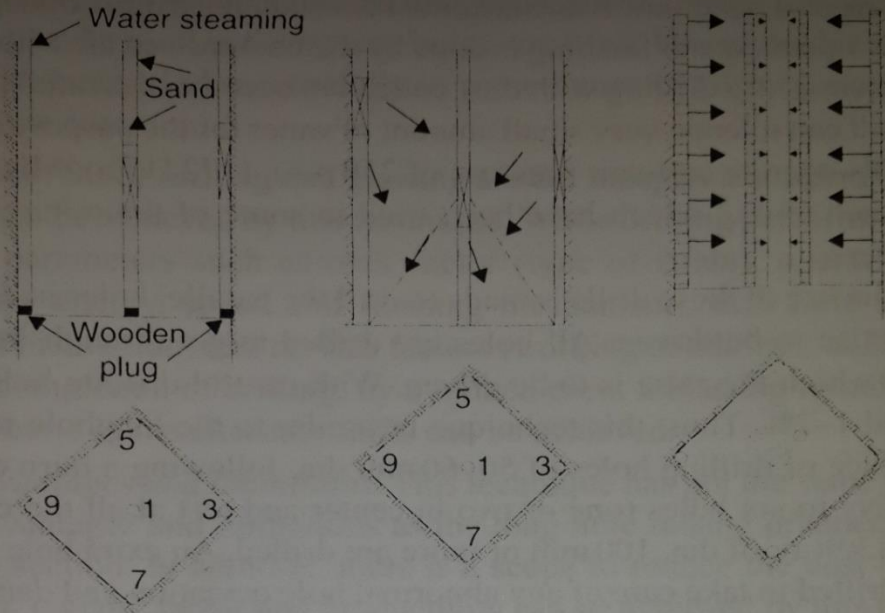
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- (d) Drilling with blasthole drifters hole length 10–30 m & dia. 50–100 mm. Long-hole raising

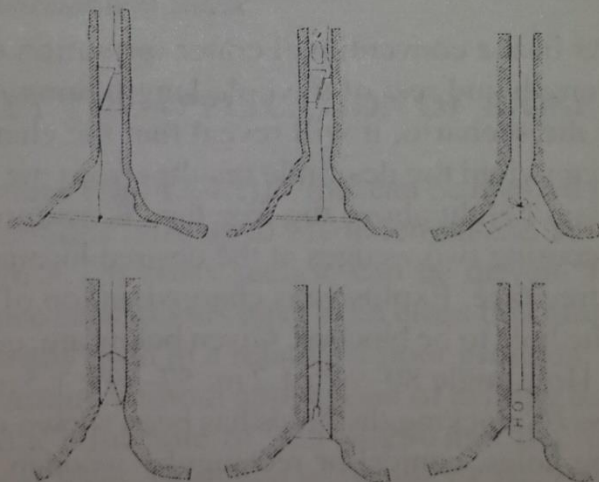
Raise Long Hole I

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(a) VCR raising – some details

(b) Conventional cylindrical cut for large-hole raising



(c) Techniques to plug bottoms of big-blast holes

Advantage I

- *Method is safe, no human exposure inside raise, drilling, blasting is performed from upper level only.
- *Raising method is fast, it is not a cyclic method.
- *Machine utilization is high, drilling is independent of raise excavation.
- *Drilling for raise can be done well in advance.
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Advantage II

- *Method is simple, no elaborate preparation & arrangement are required for raising.
- *This method of raising is the cheapest.

Disadvantages I

- *Long hole blasting technique is possible, if both levels (upper level & lower level) are accessible.
- *Raises up to 40m is possible to excavate by this method.
- *Raises in disturbed ground is difficult by this method.
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Vertical Crater Retreat (VCR) I

Vertical Crater Retreat (VCR) I

- *In long hole raise are also excavated by “vertical crater retreat” blasting method.
- *Crater is a cavity created by explosive charged in hole with only one free face.
- *In raising method crater are formed down wards to break rock around hole.
- *Since craters are facing down, immediately after blasting, all broken material falls down.
- *Because of blasting of explosive in hole, a stress zone of elliptical shape is formed around hole.

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Vertical Crater Retreat (VCR) II

- *Exact shape, size & zone of influence depends on stability of rock, rock properties, geological structure etc.
- *Length to Diameter ratio also depends up on stability of rock, rock properties, geological structure.
- *Raise up to 150m long can be excavated by VCR method.

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VCR Blasting I

*In blasting, spherical charge is defined as length (L) to diameter (D) ratio.

L> length of explosive column in blast hole.

D> diameter of blast hole.

*Normal ratio for L:D is 4:1 but not exceeding 6:1.

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VCR Blasting Formula I

$$(1) L = 6 \times d, \quad (2) L_o = 0.5 \times L_c, \quad (3) L_c = S \times Q^{1/3}$$
$$(4) Q = 3 \times d^3 \times \pi \times D / 2 \quad (4) L_o = 0.5 \times S \times Q^{1/3}$$

L> Explosive charge length mm.

L_o> optimum explosive column length mm.

L_c = critical explosive column length mm.

d> diameter of blast hole mm

S> strain energy factor, usually 1.5 but depends upon explosive & type of rock.

Q> explosive weight kg.

D> explosive density gm/cc

VCR Drilling I

- *In general, drilling system is same as that of long hole raise except following changes.
- *Drill hole diameter is 100mm to 165mm diameter, 165mm is common diameter.
- *In VCR system of raising, 5 holes are drilled, 1 center hole & 4 holes at 4 corners.
- *An extra hole is drilled at center to take care of abnormal deviation or damage of hole while blasting.

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VCR Hole Plugging I

- *Unlike long hole raising, in VCR method, explosive columns are to be kept at different height for yielding desirable blasting performance.
- *Different holes are to be plugged at different height from bottom.
- *Hole blocking begins by tying one wedge block on to polypro-pilin rope.
- *Wedge is lowered down to pre determined blocking height.
- *Wedge block is suspended from top of hole with help of 5-6mm polypro-pilin rope.

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VCR Hole Plugging II

- *Wedge block is placed at location where blockage has to take place.
- *Another wedge is lowered to block blast hole at desirable place.
- *After second wedge block is lowered down so that it hits first one.
- *Polypropylene rope is tugged to ensure that block wedges together against wall of hole.

VCR Hole Plugging III

- *About 30cm of rock cutting is powered in hole for proper blocking.
- *Any extra hole, which is not charged, should be filled with stemming material to avoid jamming at time of blasting raise round.

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VCR Blasting I

- *Holes are loaded with explosive from top.
- *Amount of explosive depends up on density of explosive & ratio of hole diameter to length of hole so that spherical charges can be obtained.
- *At first half of weight of explosive is lowered in hole.
- *Then primer with proper delay is lowered in to hole.
- *Rest of explosive is lowered in hole.
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VCR Blasting II

- *Hole is stemmed for a length of 1 to 2m, using suitable material.
- *Same process is followed for all holes.
- *Low density explosive like ANFO is not suitable for VCR method.
- *Only high density explosive like emulsion explosive is used in VCR technique.

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VCR Blasting III

- *Use of double deck or multiple deck charging has been tried for VCR blasting.
- *Multi deck charging is more common in stoping operation.
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Advantage I

- *VCR raising method has all advantages of long hole raising.
- *Additional advantages
- *Cost of raising comes down further, due to less number of holes, longer blasting round, higher powder factor.
- *Raising up to 150m long can be excavated by VCR.
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Disadvantage I

*In addition to disadvantages of long hole raising, VCR method of raising involve higher capital expenditure for purchasing higher capacity drill machine.

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Reference I

*Surface and Underground Excavations – Methods, Techniques and equipment. Ratan Raj Tatiya, Page 340 to 347.